

Recover deleted or damaged files from a storage device using TestDisk

EXPERIMENT NO: 02

DATE:

1. AIM

To recover deleted files and repair a missing or corrupted partition from a storage device using the TestDisk data recovery command-line utility.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** TestDisk (v7.3-WIP Data Recovery Utility)
- **Hardware:** Target Storage Device (e.g., SanDisk cruzer Blade16GB USB Device).

3. DESCRIPTION

TestDisk is a powerful, open-source data recovery software primarily designed to help recover lost partitions and make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses, or human error (such as accidentally deleting a Partition Table). It can also be utilized to undelete files from FAT, exFAT, NTFS, and ext2 file systems. TestDisk queries the BIOS or the OS in order to find the hard drives and their characteristics, performs a quick check of the disk structure, and looks for lost partitions or deleted files.

4. PROCEDURE

Step 1: Log Creation & Disk Selection

1. Open the testdisk_win.exe application as an Administrator.
2. **Log Creation:** Choose [Create] to instruct TestDisk to create a log file containing technical information and messages. Press Enter to proceed.

```
E:\USB_Evidence_01\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything
```

Figure 1: Initial TestDisk dashboard prompting for log creation

3. **Disk Selection:** All connected hard drives will be detected and listed. Use the up/down arrow keys to select the target hard drive containing the lost partitions or deleted files (e.g., Disk \\.\PhysicalDrive2 - SanDisk cruzer Blade16GB USB Device) Press Enter to Proceed.

```
E:\USB_Evidence_01\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 1000 GB / 931 GiB - WDC WD10SPZX-75Z10T3
Disk \\.\PhysicalDrive1 - 240 GB / 223 GiB - WDC WDS240G2G0B-00EPW0
>Disk \\.\PhysicalDrive2 - 16 GB / 14 GiB - SanDisk Cruzer Blade

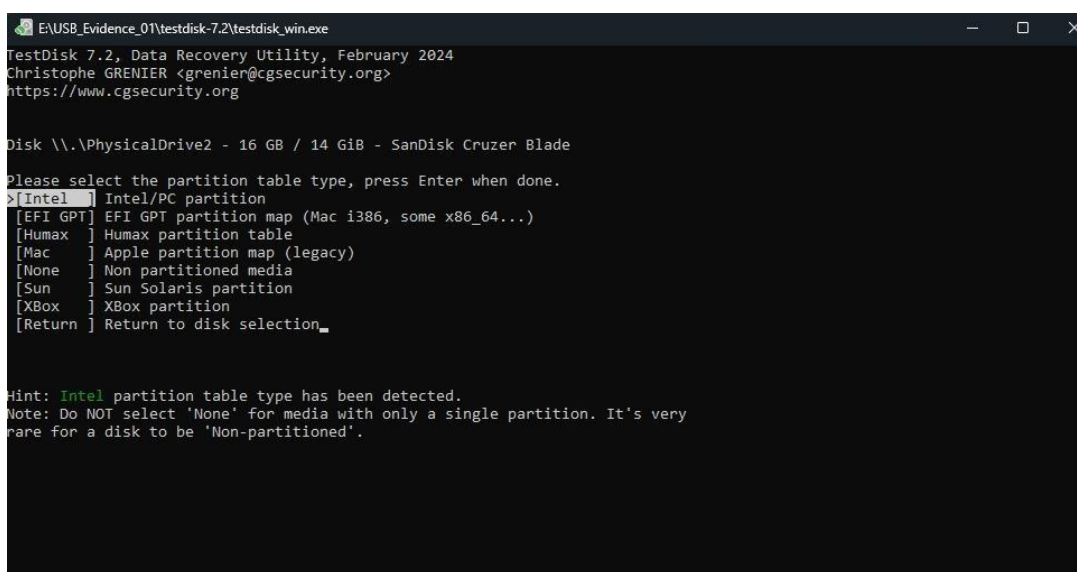
> [ Proceed ] [ Quit ]

Note: Serial number 4C530399970324123381
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers
```

Figure 2: Selecting the physical storage drive in TestDisk

Step 2: Partition Table Type Selection

1. TestDisk displays the available partition table types.
2. Select the partition table type. Usually, the default value is the correct one as TestDisk auto-detects the partition table type (e.g., [Intel] Intel/PC partition). Press Enter to Proceed.



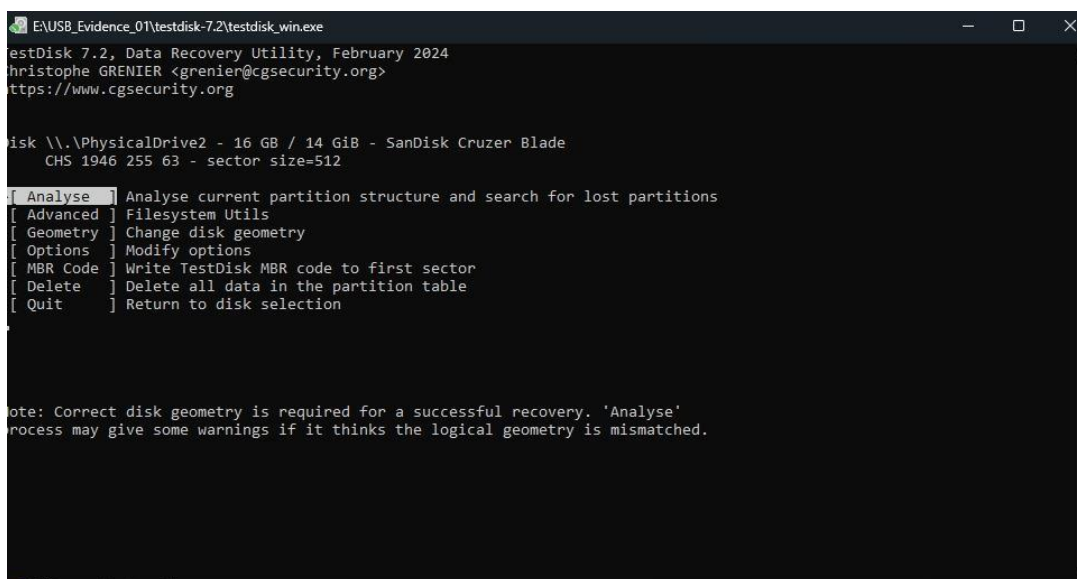
```
EAUSB_Evidence_01\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 16 GB / 14 GiB - SanDisk Cruzer Blade

Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection_

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

Figure 3: Selecting the Intel/PC partition table type



```
EAUSB_Evidence_01\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 16 GB / 14 GiB - SanDisk Cruzer Blade
CHS 1946 255 63 - sector size=512

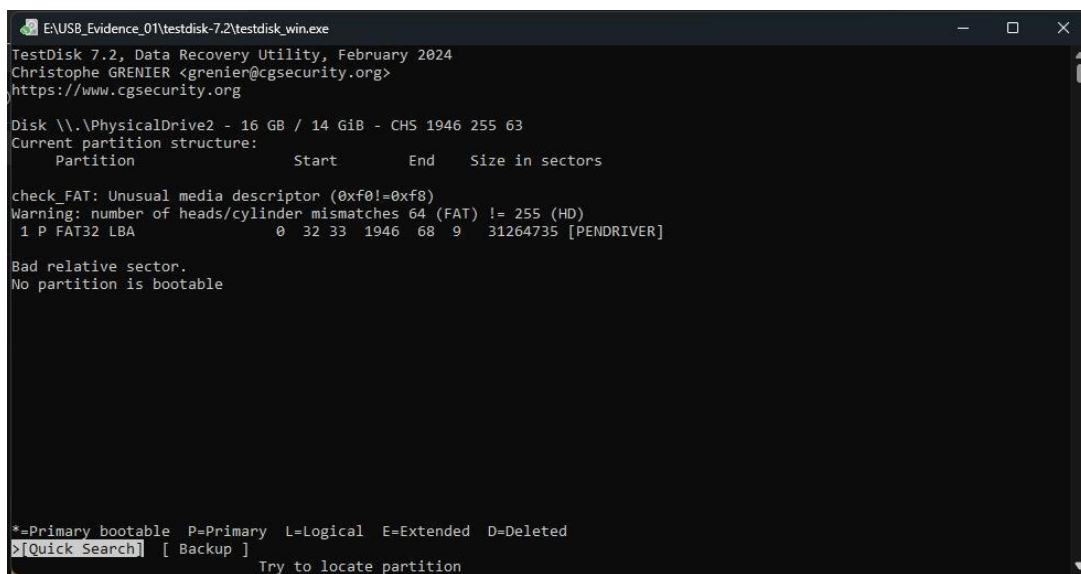
[Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[MBR Code] Write TestDisk MBR code to first sector
[Delete] Delete all data in the partition table
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

Figure 4: Choosing the Analyse option to examine the partition structure

Step 3: Current Partition Table Status

1. TestDisk displays the main menu options. Use the default menu [Analyse] to check your current partition structure and search for lost partitions. Confirm at Analyse with Enter.
2. The current partition structure is listed. Examine the structure for missing partitions and errors (e.g., Invalid NTFS boot, missing logical partitions). Confirm at [Quick Search] to proceed.



```
E:\USB_Evidence_01\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive2 - 16 GB / 14 GiB - CHS 1946 255 63
Current partition structure:
  Partition          Start          End      Size in sectors
check_FAT: Unusual media descriptor (0xf0!=0xf8)
Warning: number of heads/cylinder mismatches 64 (FAT) != 255 (HD)
 1 P FAT32 LBA        0 32 33 1946 68 9 31264735 [PENDRIVER]

Bad relative sector.
No partition is bootable

*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [ Backup ]
Try to locate partition
```

Figure 5: Initiating a Quick Search for lost partitions

Step 4: Deeper Search (If Required)

1. If the quick search does not find the required partition or if partitions overlap (status 'D' for deleted), highlight the [Deeper Search] menu and press Enter.
2. Deeper Search will scan each cylinder for FAT32 backup boot sectors, NTFS backup boot superblocks, etc., to detect more partitions.
3. Review the discovered partitions, change their status if necessary (using the arrow keys), and press **Enter** to proceed to the next screen.

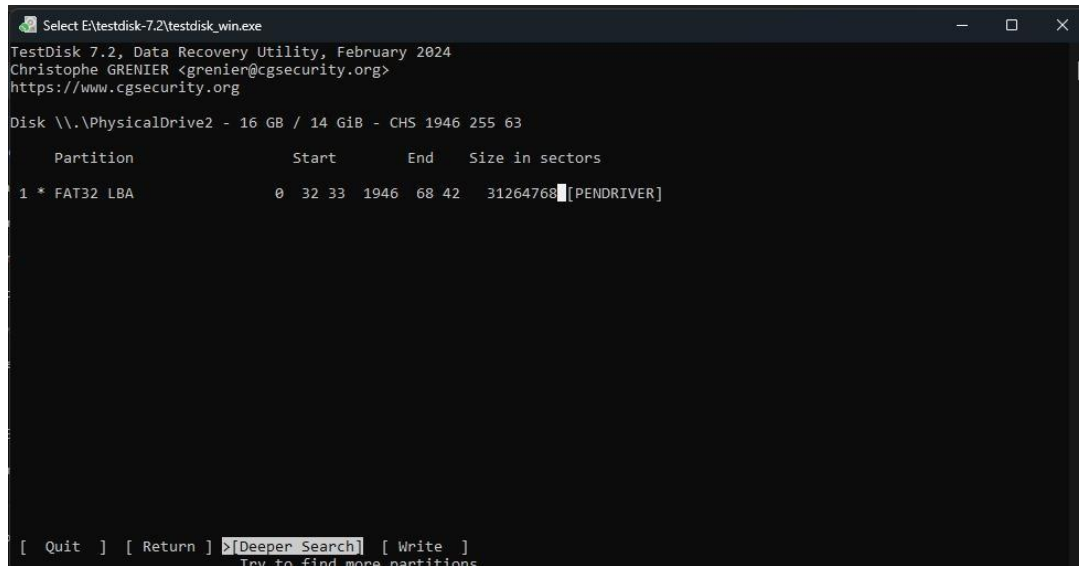


Figure 6: Selecting the Deeper Search option

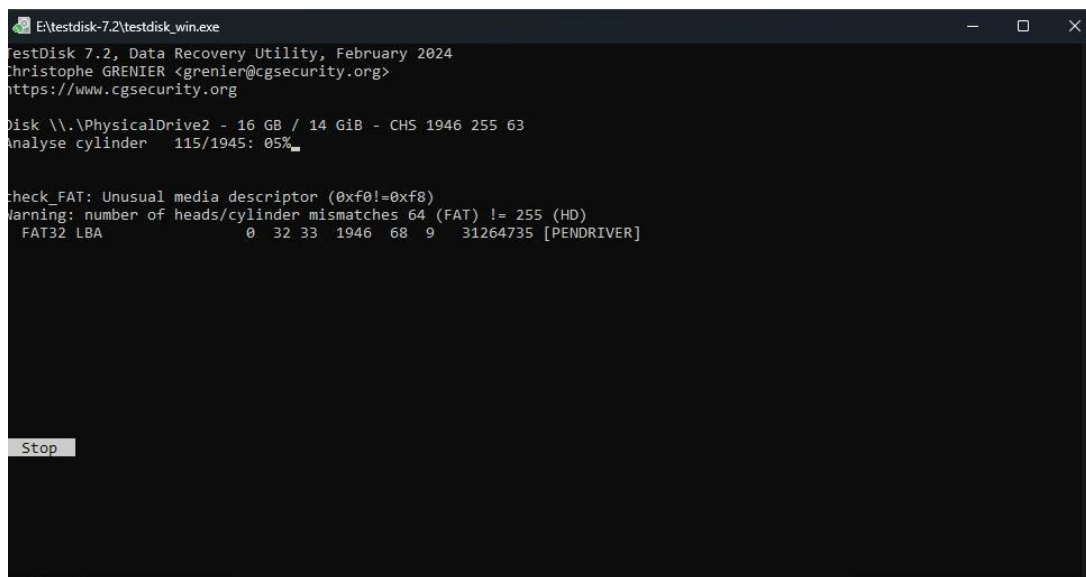


Figure 7: Deeper Search actively scanning disk cylinders

Step 5: Identifying and Accessing the Correct Partition

1. Once the search is complete, highlight the correct partition you wish to recover or extract files from (e.g., >L FAT32 LBA).
2. Press the p key to list the files inside that partition.

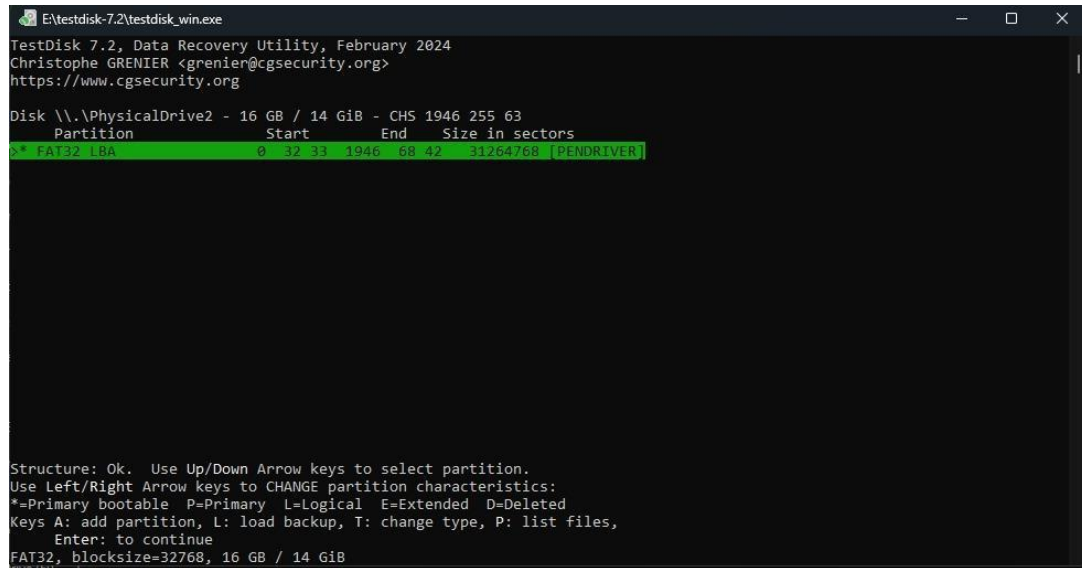


Figure 8: Highlighting the located FAT32 partition

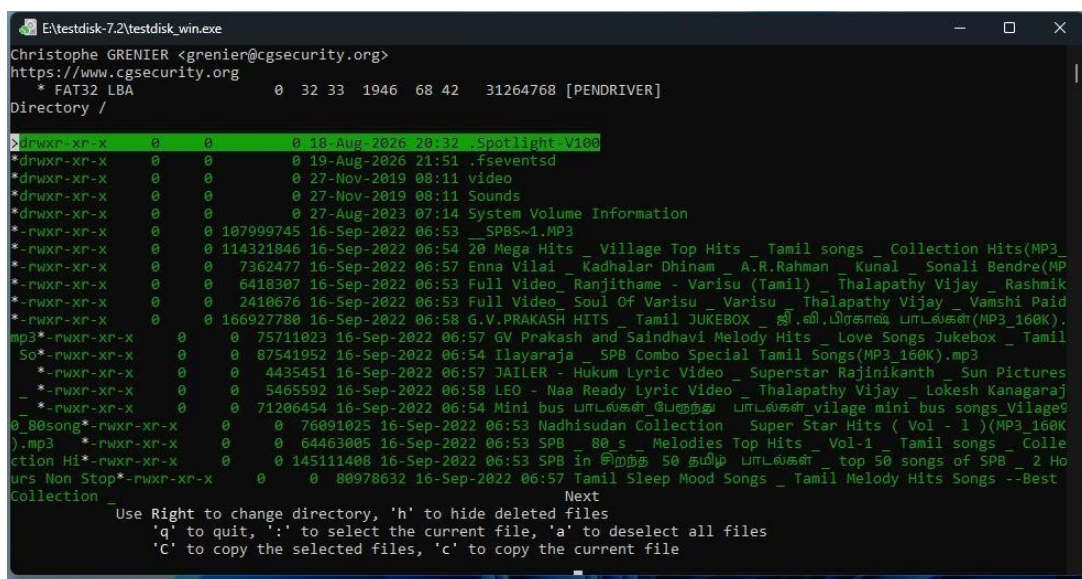


Figure 9: Listing the directory contents and selecting a file for recovery

Step 6: Recovering / Copying Deleted Files

1. In the file listing, navigate through the folders using arrow keys. Files listed in red indicate deleted entries.
2. Highlight the specific deleted file you want to recover (e.g., images (1).jpg).

-
- ```

E:\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024

Please select a destination where the marked files will be copied.
Keys: Arrow keys to select another directory
 C when the destination is correct
 Q to quit

Directory E:\testdisk-7.2
>drwxr-xr-x 197609 197609 0 26-Aug-2026 10:40 ..
drwxr-xr-x 197609 197609 0 1-Jan-1980 00:00 .
drwxr-xr-x 197609 197609 0 26-Aug-2026 10:41 63
drwxr-xr-x 197609 197609 0 26-Aug-2026 10:41 platforms
-rw-r--r-- 197609 197609 220 1-Jan-1980 00:00 AUTHORS.txt
-rw-r--r-- 197609 197609 18326 1-Jan-1980 00:00 COPYING.txt
-rw-r--r-- 197609 197609 117 1-Jan-1980 00:00 INFO
-rw-r--r-- 197609 197609 20971 1-Jan-1980 00:00 NEWS.txt
-rwxr-xr-x 197609 197609 6324655 1-Jan-1980 00:00 Qt5Core.dll
-rwxr-xr-x 197609 197609 6921855 1-Jan-1980 00:00 Qt5Gui.dll
-rwxr-xr-x 197609 197609 7303411 1-Jan-1980 00:00 Qt5Widgets.dll
-rw-r--r-- 197609 197609 350 1-Jan-1980 00:00 THANKS.txt
-rw-r--r-- 197609 197609 37 1-Jan-1980 00:00 VERSION.txt
-rwxr-xr-x 197609 197609 5457751 1-Jan-1980 00:00 cygwin-2.dll
-rwxr-xr-x 197609 197609 491530 1-Jan-1980 00:00 cyggcc-s_seh-1.dll
-rwxr-xr-x 197609 197609 1380305 1-Jan-1980 00:00 cygiconv-2.dll
-rwxr-xr-x 197609 197609 1994733 1-Jan-1980 00:00 cygjpeg-8.dll
-rwxr-xr-x 197609 197609 3785114 1-Jan-1980 00:00 cygncursesw-10.dll
-rwxr-xr-x 197609 197609 3693153 1-Jan-1980 00:00 cygwin1.dll
-rwxr-xr-x 197609 197609 306623 1-Jan-1980 00:00 cygz.dll
-rw-r--r-- 197609 197609 504 1-Jan-1980 00:00 documentation.html
-rw-r--r-- 197609 197609 210 26-Aug-2026 10:41 fidentify.log

Next

```

```
E:\testdisk-7.2\testdisk_win.exe
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* FAT32 LBA 0 32 33 1946 68 42 31264768 [PENDRIVER]
Directory /video
Copy done! 0 ok, 0 failed
drwxr-xr-x 0 0 0 18-Aug-2026 20:32 .Spotlight-V100
drwxr-xr-x 0 0 0 19-Aug-2026 21:51 .fsevents
> drwxr-xr-x 0 0 0 27-Nov-2019 08:11 video
drwxr-xr-x 0 0 0 27-Nov-2019 08:11 Sounds
drwxr-xr-x 0 0 0 27-Aug-2023 07:14 System Volume Information
```

Figure 11: Successful copy/recovery of the deleted file

### **Step 7: Partition Table and Boot Sector Recovery (Optional)**

1. Press q to quit the file listing and return to the partition selection.
2. Use the left/right arrow keys to change the status of a partition from D (Deleted) to L (Logical) or P (Primary) to recover the partition itself. Press Enter.
3. If all partitions are correctly listed, confirm at [Write] with Enter, press y to confirm, and then OK.
4. If a boot sector is damaged, use the [Backup BS] option to copy the backup boot sector over the damaged one.
5. Exit TestDisk and reboot the computer to access the restored partition data.

## **5. RESULT**

The TestDisk data recovery utility was successfully executed to analyze the physical drive's geometry and partition tables. Lost partitions were successfully identified through deeper searching, and deleted files were effectively navigated, selected, and copied (recovered) to a secure local destination.